

Variance Reduction Techniques in Practice

Jaime Romero-Barrientos

*Research Center in Neutron Spectroscopy and Nuclear Physics, Chilean Nuclear Energy Commission,
Chile*

Millenium Institute For SubAtomic Physics at High Energy Frontier, SAPHIR

CERN School of Computing 2026, Santiago, Chile



Respond at
pe.app/jaimeromero



In variance reduction, what is actually being biased?

36

11%
The underlying physical cross sections.

0%
The Monte Carlo estimator itself.

78%
The sampling probability distribution.

11%
The final expected value of the observable.



+ Add Choice

When Presented



Outline

1. From preserving the mean to optimal biasing
2. Spatial Techniques
3. Weight Windows and Weight Control
4. Physics-based biasing
5. Validation
6. Takeaways

From Preserving the Mean to Optimal Biasing

Recap: The Monte Carlo Estimator

The Equation

- We defined a General Estimator:
 - $Q = E[x] = \int h(x)f(x)dx$
 - $f(x)$: Physical PDF (Physics).
 - $h(x)$: Score function (Detector).

The Problem

- Rare events: $f(x)$ is tiny in the region of interest.
- Analog limit: $R \propto 1/\sqrt{N}$
- Result: Increasing CPU alone is not an option for rare events.

The Key Identity: Biased Sampling

- Our key result → **Biased Sampling**:

$$Q = \int h(x) \underbrace{\left[\frac{f(x)}{\hat{f}(x)} \right]}_{\text{Weight (w)}} \hat{f}(x) dx$$

- **How to proceed:**

1. Sample from the **biased PDF** $\hat{f}(x)$.

2. Correct with **weight** $w = \frac{f(x)}{\hat{f}(x)}$.

- Result: The Mean (Q) remains **unbiased**.

The Danger of a Bad $\hat{f}(x)$

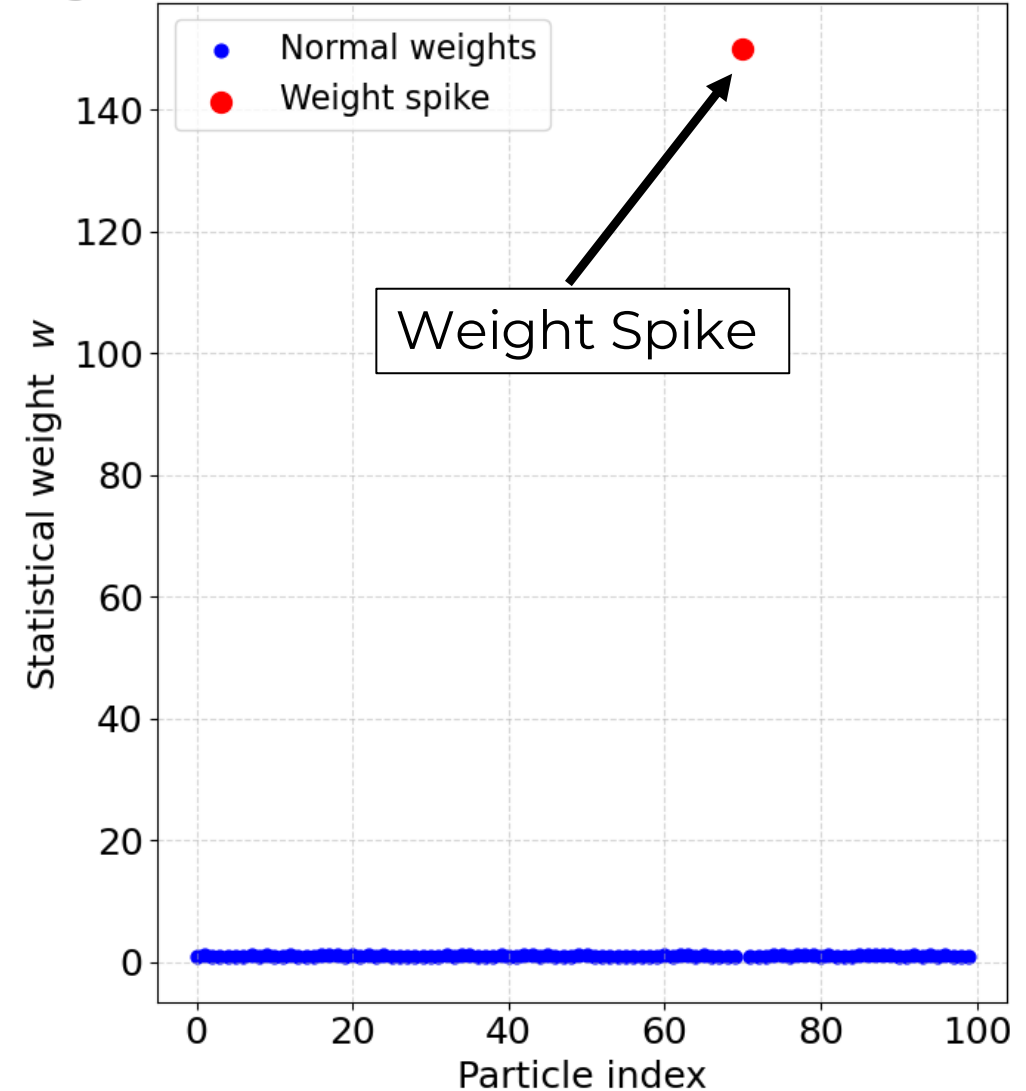
- **Weight Definition:**

$$w(x) = \frac{f(x)}{\hat{f}(x)}$$

- **Risk & Consequences:**

- If $\hat{f}(x)$ is very small where $h(x)f(x)$ is **large** → **Huge weights**.
- **“Weight spikes”**: few particles dominate the result.
- Variance can become **arbitrarily large (even infinite)**.

Weight Distribution: One Particle Dominates the Statistics

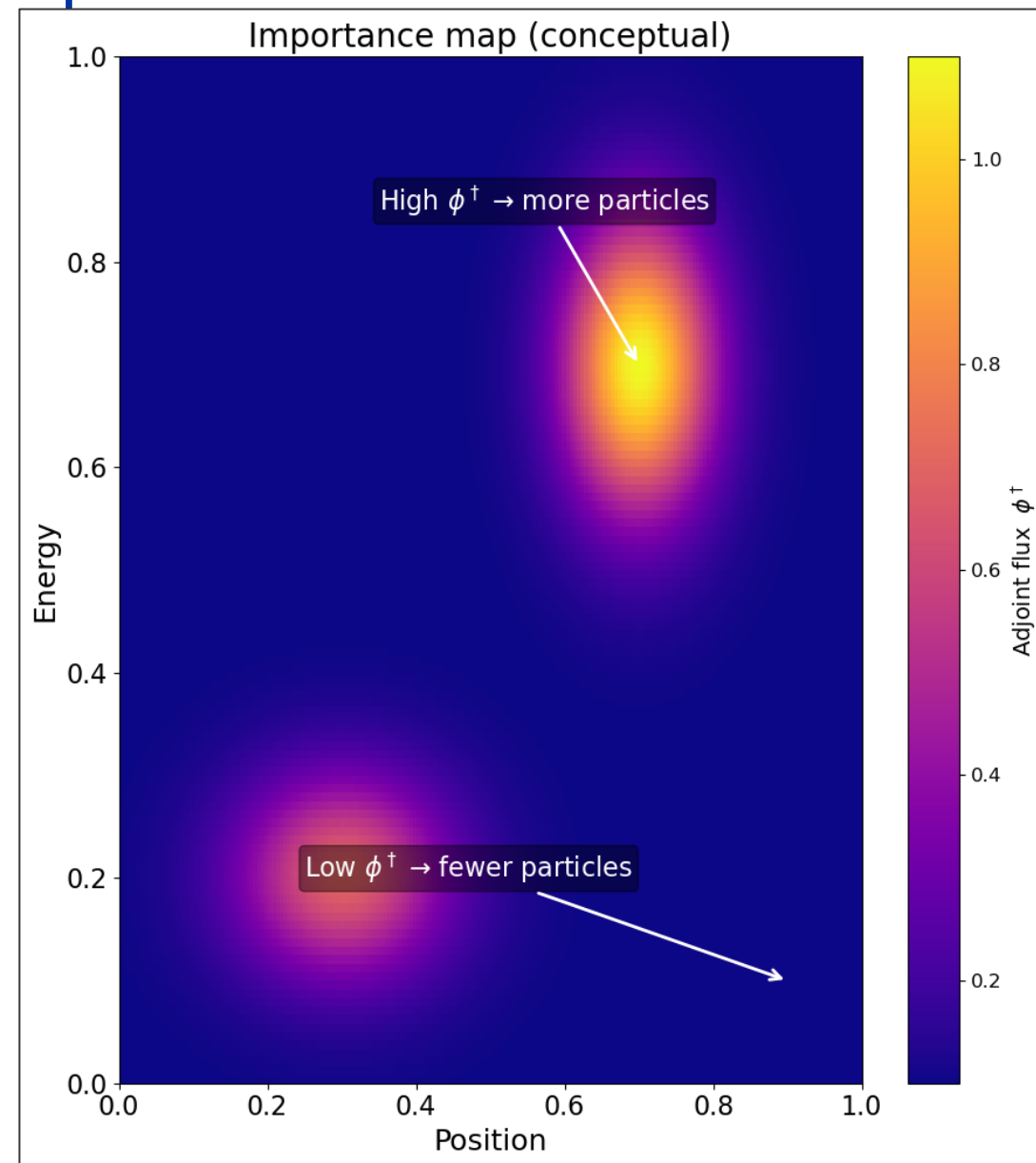


Is There a “Perfect” $\hat{f}(x)$?

- **The Optimal Biasing Function:**
 - It samples particles **exactly proportional** to their contribution.
 - $f_{opt} \propto |h(x) \cdot f(x)|$
- **The Result (Zero Variance):**
 - All histories contribute the same value.
 - Uncertainty $\sigma = 0$ with $N = 1$.
- **So... Why don't we use it?**
 - Because building f_{opt} requires **knowing $h(x)f(x)$ beforehand**.

The Goal: Approximating Importance

- **The Importance Function (Adjoint $\phi^\dagger$):**
 - $\phi^\dagger(\vec{r}, E)$ represents the expected contribution of a particle at $(\vec{r}, E)$ to the detector.
- **Practical VR:**
 - We cannot know $\phi^\dagger(\vec{r}, E)$ exactly.
 - BUT we can estimate it.
- **The Strategy:**
 - **Encourage** particles where importance is **High**.
 - **Suppress** particles where importance is **Low**.



Spatial Techniques: From Concept to Practice

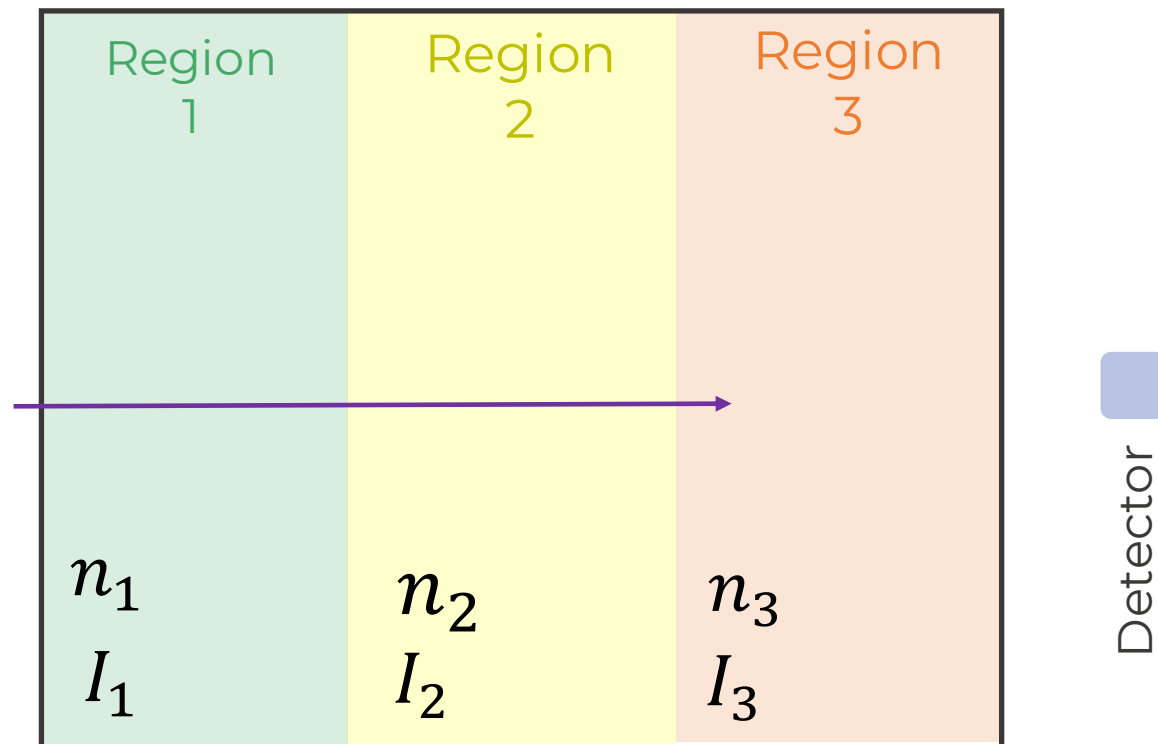
Spatial Importance

- **Recall Block 1:**

- Assign an Importance I to each geometric region.
- **I increases as we move towards the detector.**

- **Goal of Spatial VR:**

- As the particle population drops, keep it roughly constant.
- More particles in high importance regions.



- Number of particles: $n_1 > n_2 > n_3$
- Importance: $I_1 < I_2 < I_3$

Number of Particles ↓ , Importance ↑

Splitting: Generating More Histories

- **Trigger:** Particle moves from $Region_{old} \rightarrow Region_{new}$ with $I_{new} > I_{old}$.

- **Integer ratio (simple case):**

$$n = \frac{I_{new}}{I_{old}} \in \mathbb{N}$$

- **Action:**

1. Split particle into n identical copies.
2. Assign new weight:

$$w_{new} = \frac{w_{old}}{n}$$

- **Result:**

- More histories in High Importance Regions.
- Lower weight per history.

- **Total weight and Mean preserved.**

Russian Roulette: Saving CPU Time

- **Trigger:** Particle moves from.
 $Region_{old} \rightarrow Region_{new}$ with $I_{new} < I_{old}$.

- **Survival Probability:**

$$p = \frac{I_{new}}{I_{old}} < 1$$

- **Action:** Play a game of chance.
 - With probability p , **particle survives** and a new weight is assigned:

$$w_{new} = \frac{w_{old}}{p}$$

- With probability $1 - p$, **particle is killed.**

- **Result:**

- Fewer histories in Low Importance Regions.
- Higher weight for survivors, mean preserved.

Spatial Techniques as Importance Sampling

- **The link:**
 - **Splitting**: artificially increases sampling density in high-importance regions.
 - **Russian Roulette**: decreases sampling density in low-importance regions.
- **Weights:**
 - Weight updates (e.g. w/n , w/p) keep the **mean unchanged**.
 - Conceptually they play the same role as the $f/\hat{f}$ correction in importance sampling.
- **Key insight:**
 - Spatial VR $\approx$ **importance sampling in geometry** (piecewise-constant importance per cell).

Splitting Logic

```
# Particle crosses surface into new cell
ratio = l_new / l_old    # importance ratio

if ratio > 1.0:          # Splitting case
    n_split = floor(ratio)

    # Handle non-integer ratios stochastically
    if random() < (ratio - n_split):
        n_split += 1

    w_child = w_old / n_Split

    kill_particle() #replace parent by copies

    # Push split particles to the bank
    for i in range(n_split):
        bank.add_particle(copy(state), w_child)
```

This snippet shows how the splitting logic could be implemented.

Russian Roulette Logic

```
# Particle crosses surface into new cell
ratio = l_new / l_old      # importance ratio

if ratio < 1.0:            # Russian Roulette case
    p_survive = ratio      # survival probability

    if random() < p_survive:
        # Particle survives: increase weight to keep mean unchanged
        w_new = w_old / p_survive
        bank.add_particle(copy(state), w_new)
    else:
        # Particle is killed: no push to the bank
        kill_particle()
```

This snippet shows how the RR logic could be implemented.

Pitfalls of Spatial Splitting

- **Particle population explosion:**
 - Very large importance jumps (e.g. $I = 1 \rightarrow I = 100$) can create **huge numbers of particles in one step.**
 - **Risk:** CPU and memory blow-up.
- **Diminishing returns due to correlations:**
 - Many nearly identical split histories add much less information than you'd expect.
 - **More particles $\neq$ always much better statistics.**
- **Need for regulation:**
 - We must keep both weights and population under control.
 - This motivates weight control / weight windows (next section).

Weight Windows and Weight Control

The Instability of Weights

- **The problem:**

- Combined techniques (splitting / RR, implicit capture, physics biasing) can make **particle weights fluctuate over many orders of magnitude.**

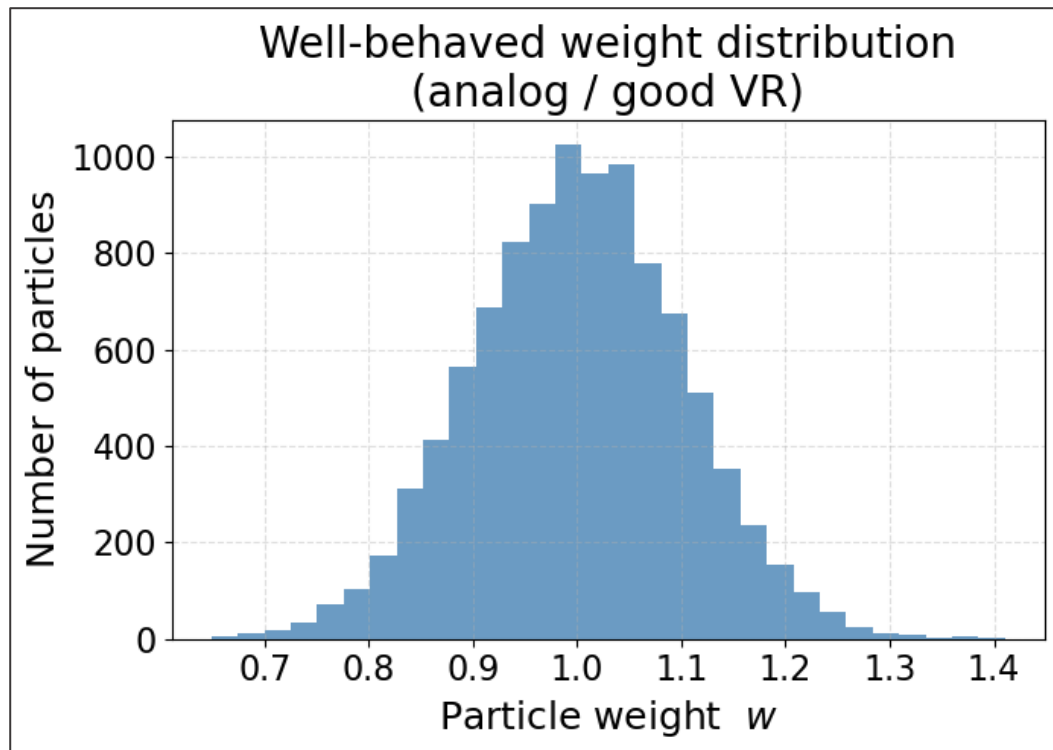
- **Weight spikes:**

- Example: one particle reaches the detector with $w = 1000$, while typical weights are $w \approx 1$.
- The average becomes heavily dominated by a **few high-weight histories** → **large variance.**

- **Need:**

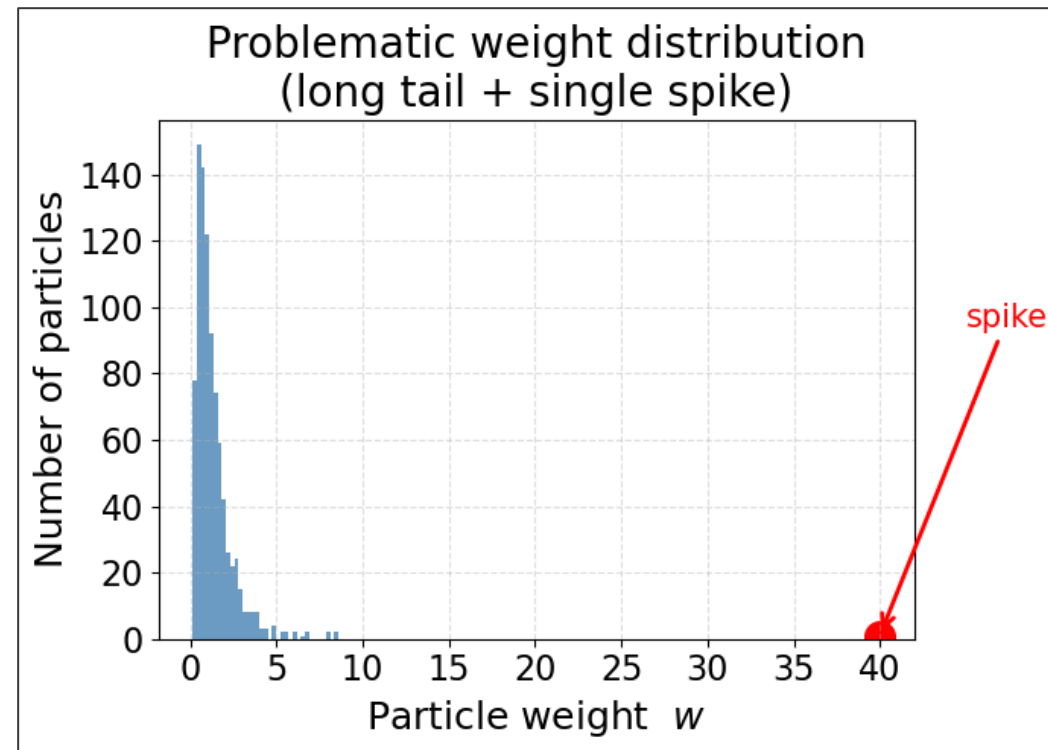
- We need a mechanism to **keep weights in a reasonable range.**

Visualizing Weight Stability



- **Analog / Good VR**

- Analog: weights concentrated at $w = 1$.
- Good VR: narrow distribution around a target weight.



- **Bad VR (Spikes)**

- Uncontrolled VR: broad weight distribution with long tails.
- Tail: rare high-weight particles (“variance killers”).

Weight Windows: The Concept

- **Definition:**

- Limit particle weight in each region / energy bin.

- **Parameters per region/energy:**

- W_L : lower weight bound.
- W_U : upper weight bound.
- W_{target} : desired weight (often between W_L and W_U).

- **Weight Control Rule:**

- If $w < W_L \rightarrow$ apply Russian Roulette.
- If $w > W_U \rightarrow$ apply Splitting
- If $W_L \leq w \leq W_U \rightarrow$ do nothing.

- **Link to Importance:**

- High importance $\rightarrow$ lower W_{target} (more particles, lighter)
- Low importance $\rightarrow$ higher W_{target} (fewer particles, heavier)

Weight Window Algorithm

1. Particle enters a new region / energy bin.
2. **Check its weight w against the window $[W_L, W_U]$.**
3. **If $w > W_U$ (too heavy):**
 - Split into n copies so that each child has $w_{child} \approx W_{target}$.
4. **If $w < W_L$ (too light):**
 - Play Russian Roulette with survival probability $p = w/W_{target}$.
 - If it survives, **set $w_{new} = W_{target}$** ; otherwise, kill.
5. **If $W_L \leq w \leq W_U$:**
 - Do nothing.

```
# Called when particle enters a new region/energy
if w > W_U:
    # Splitting branch
    n_split = ceil(w / W_target)
    w_child = w / n_split
    for i in range(n_split):
        bank.add_particle(copy(state), w_child)

elif w < W_L:
    # Russian Roulette branch
    p_survive = w / W_target
    if random() < p_survive:
        w_new = W_target
        bank.add_particle(copy(state), w_new)
    else:
        # particle killed
        kill_particle()
else:
    # Inside the window: no action
    bank.add_particle(copy(state), w)
```

Impact on FOM

- **Killing low weights ($w < W_L$):**
 - **Reduces CPU spent on histories with tiny expected contribution.**
 - For the same statistical quality, runtime T usually decreases (or does not grow as fast).
- **Splitting high weights ($w > W_U$):**
 - Spreads very large weights into several moderate ones.
 - **Reduces the impact of “weight spikes” on the variance**, so σ^2 often decreases.
- **Net effect (with well-designed windows):**
 - Figure of Merit **$FOM = \frac{1}{\sigma^2 T}$ typically increases** vs. analog / naive VR.

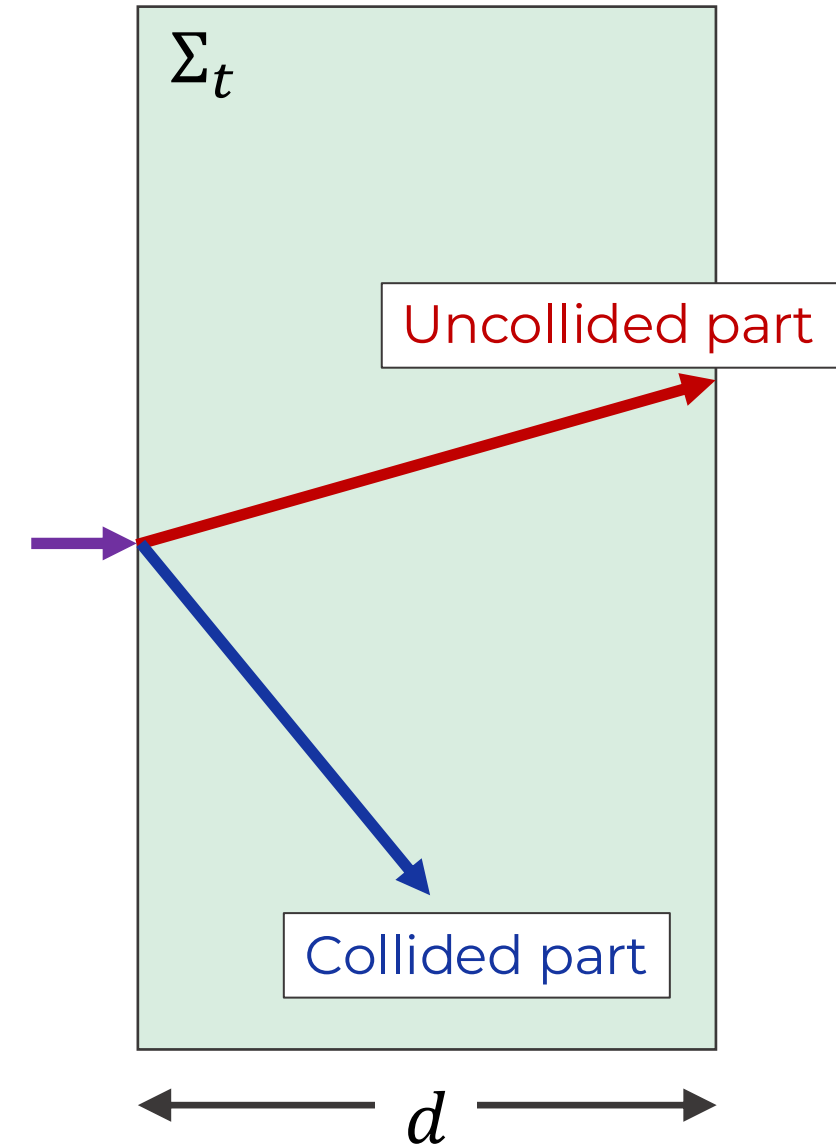
Physics-Based Biasing

Physics-Based Biasing: Beyond Space

- **Spatial VR:**
 - Controls particle population between collisions (splitting / RR / weight windows).
- **Physics-based biasing:**
 - **Modifies how collisions (interactions) happen:**
 - force interactions in thin targets,
 - enhance transmission in deep shields,
 - increase sampling of rare reaction channels.
- **Goal:**
 - Spend **more histories on rare but important events**,
 - while keeping the **mean unbiased**.

Example: Forced Collisions

- **Scenario:**
 - Thin target with $P_{int} \approx \Sigma_t \cdot d \ll 1$.
- **Analog behaviour:**
- Collision probability $P_{int} = \Pr(s \leq d) = \int_0^d \Sigma_t e^{-\Sigma_t s} ds = 1 - e^{-\Sigma_t d}$
 - Most particles cross without interacting $\rightarrow$ poor statistics.
- **Forced Collisions** $\rightarrow$ Represent both outcomes in one history:
 - **uncollided part (no interaction)**
 - **collided part (interaction in the target)**, each with appropriate weight.
- **Goal:**
 - Increase the number of effective “collision samples” without biasing the mean.



Forced Collisions: The Algorithm

- **Thin cell:** chord length d , total cross section Σ_t .

- **True probabilities:**

$$P_{no} = e^{-\Sigma_t d}, \quad P_{coll} = 1 - e^{-\Sigma_t d}$$

- Split one history (**weight w_0**) into:

- **Uncollided branch:**

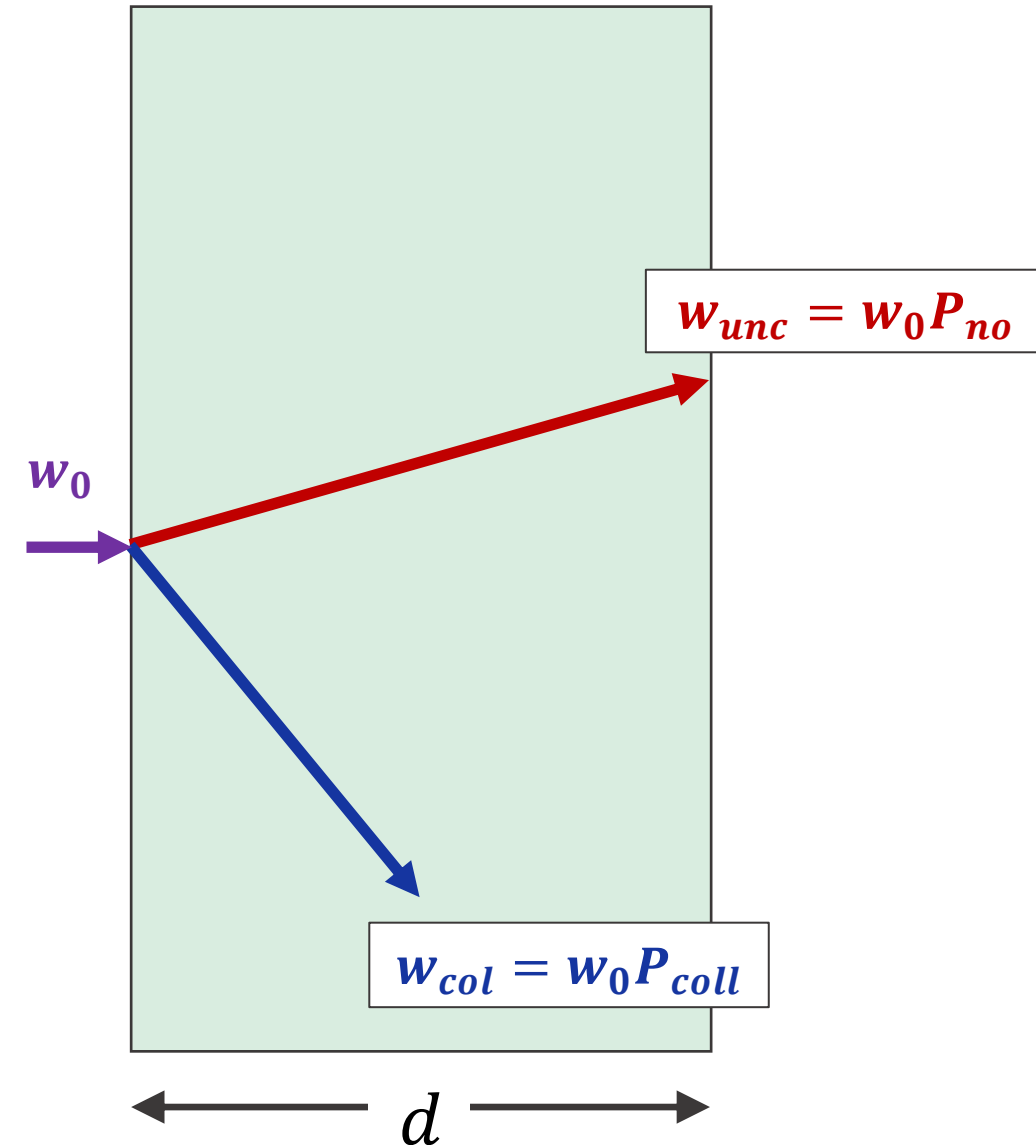
$$w_{unc} = w_0 P_{no} \rightarrow \text{particle at cell boundary.}$$

- **Collided branch:**

$$w_{col} = w_0 P_{coll} \rightarrow \text{forced collision in the cell.}$$

- **Collision point:**

- Distance $s \in [0, d]$ sampled from an exponential truncated to $[0, d]$ for **collided branch**.



Example: Exponential Transform

- **Scenario:**

- Deep shielding, flux drops over many mean free paths.

- **Biasing:**

- **Reduce the effective total cross section in the forward direction**

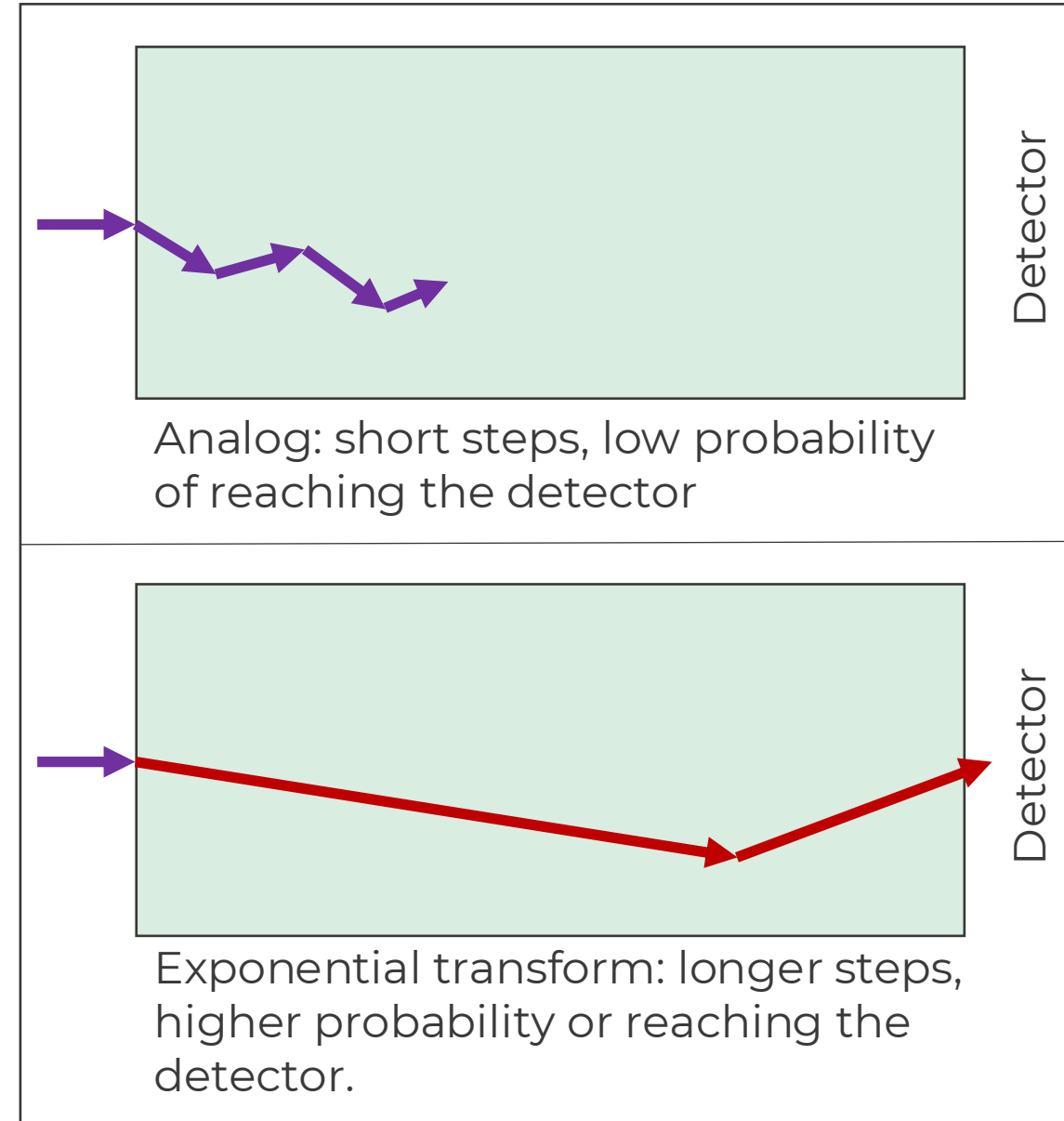
$$\Sigma_{biased} = \Sigma_t(1 - p\mu), \quad 0 \leq p < 1 \quad (\mu: \text{cosine toward the detector})$$

- **Effect:**

- **Particles travel further between collisions along $+\mu$.**

- **Weight update (per flight of length s):**

$$w_{new} = w_{old} \frac{\Sigma_t}{\Sigma_{biased}} \exp[-(\Sigma_t - \Sigma_{biased})s]$$



Example: Survival Biasing

- **Scenario:**

- Capture-dominated material.

- **Analog Behavior:**

- Many tracks are killed by absorption near the surface.

- **Biasing idea:**

- Never kill the track on capture.
- Treat absorption as weight loss, particle always survives.

- **Weight correction per collision:**

- New weight (surviving part):

- $w_{new} = w_{old} \frac{\Sigma_s}{\Sigma_t}$

- Absorbed part (to score the captures):

- $\Delta w_{abs} = w_{old} \frac{\Sigma_a}{\Sigma_t}$

Physics Biasing: Use Cases

- **Forced Collisions**
 - Thin targets / detector foils.
 - Very low interaction probability (e.g. neutrino-like rare-event problems).
- **Exponential Transform**
 - Deep shielding, many mean free paths.
 - When geometry splitting + weight windows are not enough.
- **Survival Biasing (Implicit Capture)**
 - Highly absorbing materials / capture-dominated regions.
 - Want both absorption tallies and flux deeper in the system.

Validation

Closing the Loop: Validation

- **The Big Question:**
 - *Is my Monte Carlo result still physically correct?*
- **Theory says “yes” if:**
 - **weights implement $w = f/\hat{f}$ correctly, and**
 - **the relevant phase space is properly sampled.**
- **Practice says “be careful”:**
 - aggressive VR can give **small error bars on a biased result.**
 - VR is “expert mode”: **easy to converge to the wrong answer.**

Validation Checklist: Accuracy

- **Compare with analog:**
 - Run a short analog case.
 - Biased mean should agree within uncertainties.
- **Compare with physics / benchmarks:**
 - Does the result follow known trends?
 - (e.g. flux attenuation slope, simple 1D model, benchmark data)
- **Sanity of global behaviour:**
 - Reaction / leakage / absorption balances remain consistent.

Validation Checklist: Efficiency

- **Figure of Merit:**

$$FOM = \frac{1}{\sigma^2 T}$$

- **Test:**

- **FOM_{biased} should be higher than FOM_{analog} for the tallies you care about.**

- **If not:**

- VR is adding **complexity without real gain.**

Validation Checklist: Stability

- **Weight distribution:**

- Inspect weight histogram at the detector.
- **No extreme spikes:** max weight not orders of magnitude above typical.

- **Error vs histories:**

- Relative error R **should decrease roughly like $1/\sqrt{N}$.**
- Sudden jumps in R suggest rare, high-weight events.

- **Contribution spread:**

- Scoring/Results **should not be dominated by a tiny number of histories.**

Respond at
pe.app/jaimeromero

You run a variance-reduced Monte Carlo simulation. Compared to an analog run: the estimated mean value agrees within uncertainties, the statistical error is significantly smaller, and the runtime was significantly reduced.

Based in this information alone, would you trust your result?

↑ 29

10%
Yes. Agreement with the analog result and smaller uncertainty are sufficient.

59%
No. Agreement and small error are encouraging, but additional consistency checks are still required.

3%
No. Variance reduction inevitably introduces bias.

28%
Yes. Variance reduction always improves reliability once validated against an analog run.



+ Add Choice

When Presented

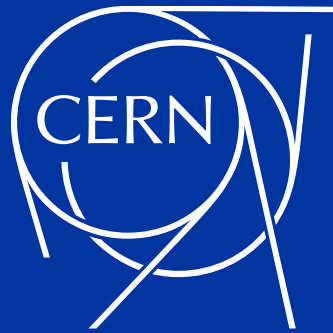


Takeaways

Key Takeaways & Next Steps

- **VR = change sampling, not physics.**
 - Bias the PDF, correct with weights, keep the mean.
- **Importance guides everything.**
 - Spatial VR and weight windows $\approx$ approximating the adjoint.
- **Physics-based biasing is powerful.**
 - Forced collisions, exponential transform, survival biasing.
- **Next step: explore & test.**
 - Apply VR Techniques in hands-on session!.

Thank you for your attention!



Questions?